

LIST OF EXPERIMENTS

- An experiment is a scientific procedure that is carried out to test a hypothesis, observe a phenomenon, or measure a variable.
- Experiments usually involve manipulating one or more independent variables and measuring their effects on one or more dependent variables.
- Experiments can be classified into different types based on their design, purpose, or outcome, such as:
 - Controlled experiments: These are experiments where all the variables except the independent variable are kept constant or controlled, and the effects of the independent variable on the dependent variable are observed.
 - Randomized experiments: These are experiments where the subjects or units are randomly assigned to different groups or treatments, and the outcomes are compared across the groups or treatments.
 - Natural experiments: These are experiments where the independent variable is not manipulated by the experimenter, but varies naturally due to some external factor, and the effects of the independent variable on the dependent variable are observed.
 - Quasi-experiments: These are experiments where the independent variable is not randomly assigned, but is determined by some pre-existing condition, and the effects of the independent variable on the dependent variable are observed.
 - Field experiments: These are experiments that are conducted in a natural setting, rather than in a laboratory or a controlled environment.
 - Laboratory experiments: These are experiments that are conducted in a controlled environment, such as a laboratory, where the variables can be manipulated and measured more precisely.
 - Simulation experiments: These are experiments that are conducted using a computer model or a mathematical equation, rather than using real subjects or units, to study the behavior of a system or a phenomenon.
 - Single-subject experiments: These are experiments that involve only one subject or unit, and the effects of different treatments or conditions are measured over time.
 - Factorial experiments: These are experiments that involve two or more independent variables, and the effects of each variable and their interactions on the dependent variable are measured.
 - Repeated measures experiments: These are experiments where the same subjects or units are exposed to different treatments or conditions over time, and the effects of the treatments or conditions on the dependent variable are measured.
 - Cross-sectional experiments: These are experiments where the sub-

- jects or units are measured at one point in time, and the differences or relationships between the variables are analyzed.
- Longitudinal experiments: These are experiments where the subjects or units are measured at multiple points in time, and the changes or trends in the variables are analyzed.

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Some possible responses for the next user turn are:

- What is the purpose of this topic?
- How many experiments do you need to perform?
- What are the criteria for selecting the experiments?
- Do you have any specific experiments in mind?

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of hardware based experiments. Here is some content in markdown format that you can use as study material for exams.

Hardware based experiments

Hardware based experiments are experiments that involve the use of physical devices, components, or systems to test a hypothesis, measure a phenomenon, or demonstrate a concept. Hardware based experiments can be classified into different types, such as:

- **Simulation experiments:** These are experiments that use software or hardware models to mimic the behavior of a real system or environment. Simulation experiments can be useful for testing scenarios that are difficult, expensive, or dangerous to perform in reality, such as natural disasters, space exploration, or nuclear reactions. Simulation experiments can also be used to study the effects of changing parameters or variables on the system or environment, such as temperature, pressure, or speed. Simulation experiments can be performed using software tools, such as MATLAB, Simulink, or LabVIEW, or hardware platforms, such as Arduino, Raspberry Pi, or FPGA.
- **Measurement experiments:** These are experiments that use sensors, instruments, or devices to collect data or information about a physical quantity, property, or phenomenon. Measurement experiments can be used to verify a theory, validate a model, or calibrate a system. Measurement experiments can also be used to explore the characteristics, behavior, or performance of a system or component, such as voltage, current, resistance, frequency, or power. Measurement experiments can be performed using tools, such as multimeters, oscilloscopes, function generators, or spectrum analyzers.

- **Design experiments:** These are experiments that involve the creation, modification, or improvement of a hardware system, component, or device. Design experiments can be used to solve a problem, meet a specification, or optimize a function. Design experiments can also be used to demonstrate the functionality, feasibility, or reliability of a system or component, such as a circuit, a robot, or a sensor. Design experiments can be performed using tools, such as breadboards, soldering irons, prototyping boards, or 3D printers.

Hardware based experiments can have various benefits, such as:

- Enhancing the understanding of theoretical concepts by applying them to practical situations.
- Developing the skills of observation, analysis, synthesis, and evaluation by performing data collection, processing, interpretation, and presentation.
- Fostering the creativity, innovation, and problem-solving abilities by designing, building, and testing hardware solutions.
- Increasing the motivation, interest, and engagement by providing hands-on, interactive, and collaborative learning experiences.

Verification of Kirchhoff's laws

Kirchhoff's laws are two rules that describe the conservation of electric current and electric potential in electrical circuits. They are named after Gustav Kirchhoff, a German physicist who formulated them in 1845. The laws are:

- Kirchhoff's current law (KCL): The algebraic sum of all currents entering and exiting a node must equal zero. This means that the current that flows into a junction of wires is equal to the current that flows out of the junction. This law is based on the principle of charge conservation, which states that electric charge cannot be created or destroyed.
- Kirchhoff's voltage law (KVL): The algebraic sum of all the voltages around any closed loop in a circuit must equal zero. This means that the sum of the voltage drops across the resistors, capacitors, inductors, and other components in a loop is equal to the sum of the voltage sources in the loop. This law is based on the principle of energy conservation, which states that energy cannot be created or destroyed.

To verify Kirchhoff's laws experimentally, we need to set up a circuit with a known voltage source, such as a battery, and measure the current and voltage at different points using an ammeter and a voltmeter. We can then apply KCL to any node in the circuit and check if the sum of the currents is zero. Similarly, we can apply KVL to any loop in the circuit and check if the sum of the voltages is zero. If the measurements agree with the laws, then we have verified them. If not, then there may be some errors in the circuit or the instruments.

2. Measurement of power and power factor in a single phase ac series inductive circuit and study improvement of power factor using capacitor

- Power factor is a measure of energy efficiency in an alternating current circuit. It is the ratio of working power (or actual power) to apparent power .
- Working power is the power consumed by the resistive part of the circuit, and apparent power is the product of the rms values of voltage and current .
- Power factor is also equal to the cosine of the phase angle between voltage and current, or the ratio of the resistance to impedance .
- Power factor ranges from 0 to 1, where 1 indicates a purely resistive circuit with no phase difference between voltage and current, and 0 indicates a purely reactive circuit with a 90 degree phase difference .
- The average power dissipated in an ac circuit is given by $P_{ave} = \frac{1}{2} I_0 V_0 \cos \phi$, where I_0 and V_0 are the peak values of current and voltage, and ϕ is the phase angle.
- A low power factor means that the circuit is drawing more current than necessary, which results in higher losses, lower efficiency, and increased costs .
- To improve the power factor, a capacitor can be connected in parallel with the inductive load. The capacitor provides a leading reactive power that cancels out some of the lagging reactive power from the inductor, thus reducing the phase angle and increasing the power factor .
- The value of the capacitor can be calculated by equating the reactive power supplied by the capacitor to the reactive power consumed by the inductor, or by using the formula $C = I / (2 \pi f V X_c)$, where I is the rms current, f is the frequency, V is the rms voltage, and X_c is the capacitive reactance .
- To measure the power and power factor in a single phase ac series inductive circuit, a wattmeter, an ammeter, and a voltmeter are connected in the circuit. The wattmeter measures the working power, the ammeter measures the rms current, and the voltmeter measures the rms voltage.
- The power factor can be calculated by dividing the wattmeter reading by the product of the ammeter and voltmeter readings, or by using the formula $\cos \phi = P / (IV)$, where P is the working power, I is the rms current, and V is the rms voltage.
- To study the improvement of power factor using a capacitor, the same circuit is used with a capacitor connected in parallel with the inductive load. The readings of the wattmeter, ammeter, and voltmeter are taken again, and the power factor is calculated using the same formula.
- The improvement of power factor can be observed by comparing the power factor values before and after connecting the capacitor. The power factor after connecting the capacitor should be higher than the power factor

before connecting the capacitor.

3. Study of phenomenon of resonance in RLC series circuit and obtain resonant frequency.

- An RLC series circuit consists of a resistor (R), an inductor (L) and a capacitor (C) connected in series to an alternating voltage source.
- The current (I) in the circuit is the same for all the elements and is given by Ohm's law: $I = V/Z$, where V is the source voltage and Z is the total impedance of the circuit.
- The impedance (Z) of the circuit is a complex quantity that depends on the frequency (f) of the source voltage and the values of R, L and C. It is given by $Z = R + j(XL - XC)$, where j is the imaginary unit, XL is the inductive reactance and XC is the capacitive reactance.
- The inductive reactance (XL) is proportional to the frequency and the inductance: $XL = 2\pi fL$. The capacitive reactance (XC) is inversely proportional to the frequency and the capacitance: $XC = 1/(2\pi fC)$.
- The phase angle (ϕ) of the circuit is the angle between the source voltage and the current. It is given by $\tan \phi = (XL - XC)/R$. The phase angle determines the power factor ($\cos \phi$) of the circuit, which is a measure of how efficiently the circuit transfers power from the source to the load.
- Resonance is a special condition in which the circuit behaves as a pure resistor, meaning that the current is in phase with the voltage and the power factor is 1. Resonance occurs when the inductive reactance is equal to the capacitive reactance: $XL = XC$. This implies that the frequency of the source voltage is equal to the resonant frequency (f_r) of the circuit, which is given by $f_r = 1/(2\pi\sqrt{LC})$.
- At resonance, the impedance of the circuit is equal to the resistance: $Z = R$. This means that the current in the circuit is maximum and is given by $I = V/R$. The voltage across the inductor and the capacitor are equal in magnitude and opposite in phase, and they cancel each other. The voltage across the resistor is equal to the source voltage.
- The quality factor (Q) of the circuit is a measure of how sharp the resonance is. It is given by $Q = XL/R = XC/R = 1/(R\sqrt{C/L})$. The higher the Q, the narrower the bandwidth (Δf) of the circuit, which is the range of frequencies around the resonant frequency for which the circuit behaves as a resonator. The bandwidth is inversely proportional to the Q: $\Delta f = f_r/Q$.

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4. Connection and measurement of power consumption of a fluorescent lamp (tube light).

- A fluorescent lamp (or tube light) is a type of electric light that uses low-pressure mercury vapor and phosphor coating to produce visible light.
- A fluorescent lamp consists of a glass tube filled with inert gas (usually argon) and a small amount of mercury, with electrodes at both ends.
- A fluorescent lamp requires a ballast to provide the necessary voltage and current to start and operate the lamp.
- A ballast can be either magnetic or electronic, depending on the type and frequency of the lamp.
- A magnetic ballast uses an iron core and a coil to create a magnetic field that induces a voltage across the lamp terminals.
- An electronic ballast uses a circuit that converts the mains voltage to a high-frequency alternating current (AC) that drives the lamp.
- A fluorescent lamp also requires a starter to initiate the discharge of the mercury vapor inside the tube.
- A starter can be either a glow switch or an electronic device, depending on the type and frequency of the lamp.
- A glow switch starter consists of a bimetallic strip and a neon lamp that heats up and bends when the lamp is switched on, closing the circuit and allowing a current to flow through the lamp.
- An electronic starter uses a capacitor and a resistor to create a pulse of voltage that triggers the lamp ignition.
- To connect and measure the power consumption of a fluorescent lamp, the following steps are required:
 - Connect the lamp, the ballast, and the starter in series with the mains supply, as shown in the figure below.
 - Use a multimeter to measure the voltage (V) across the lamp terminals and the current (I) flowing through the lamp.
 - Use the formula $P = VI$ to calculate the power (P) consumed by the lamp in watts (W).
 - Alternatively, use a wattmeter to measure the power directly.

Figure: Connection and measurement of power consumption of a fluorescent lamp

Figure: Connection and measurement of power consumption of a fluorescent lamp

- Some factors that affect the power consumption of a fluorescent lamp are:
 - The type and frequency of the lamp and the ballast, which determine the voltage and current required to operate the lamp.
 - The temperature and pressure of the gas inside the tube, which affect the resistance and conductivity of the lamp.
 - The age and condition of the lamp, which affect the efficiency and luminosity of the lamp.
 - The ambient light and humidity, which affect the starting and operating characteristics of the lamp.

5. Measurement of power in 3- phase circuit by two-wattmeter method and determination of its power factor for star as well as delta connected load.

- The two-wattmeter method is a technique for measuring the total power in a balanced three-phase circuit using two wattmeters .
- A wattmeter is an instrument that measures the power in a single-phase or balanced three-phase circuit by multiplying the voltage and current signals and averaging them over time .
- The two-wattmeter method involves connecting the current coils of the two wattmeters in series with any two line conductors, and the potential coils of each wattmeter to the third line conductor .
- The connection diagram of the two-wattmeter method for a star-connected load is shown below:

Two-wattmeter method for star-connected load

- The connection diagram of the two-wattmeter method for a delta-connected load is shown below:

Two-wattmeter method for delta-connected load

- The total power in the three-phase circuit is given by the sum of the readings of the two wattmeters, i.e., $P = P_1 + P_2$.
- The power factor of the load is given by the ratio of the total power to the product of the line voltage and the total line current, i.e., $\text{pf} = P / (V_L I_L)$.
- Alternatively, the power factor of the load can be determined by using the following formula, which involves the readings of the two wattmeters and the phase angle between the line voltage and the line current, i.e., $\text{pf} = \cos(\theta) = (P_1 + P_2) / (\sqrt{3} V_L I_L)$.
- The phase angle between the line voltage and the line current can also be calculated by using the following formula, which involves the readings of

the two wattmeters and the power factor of the load, i.e., $\cos^{-1}(\text{pf}) = \tan^{-1}(\sqrt{3} (P_2 - P_1) / (P_1 + P_2))$.

- The two-wattmeter method can be used for both balanced and unbalanced loads, but the power factor formula is valid only for balanced loads .
- The two-wattmeter method has the advantage of being simple, accurate, and economical, as it requires only two wattmeters instead of three .
- The two-wattmeter method has the limitation of being unable to measure the power in a three-phase circuit when the load is purely inductive or capacitive, as one of the wattmeters will read zero or negative .

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6. Determination of parameters of ac single phase series RLC circuit

- A series RLC circuit is a circuit that consists of a resistor (R), an inductor (L), and a capacitor (C) connected in series to an alternating current (ac) source.
- The parameters of a series RLC circuit are the resistance (R), the inductance (L), the capacitance (C), the voltage (V), the current (I), the frequency (f), the impedance (Z), the phase angle (ϕ), the power factor (pf), and the power (P).
- The voltage (V) is the peak value of the sinusoidal voltage source. The current (I) is the peak value of the sinusoidal current that flows through the circuit. The frequency (f) is the number of cycles of the ac source per second, measured in hertz (Hz).
- The impedance (Z) is the total opposition to the current flow in the circuit, measured in ohms (Ω). It is given by the formula:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

where X_L is the inductive reactance and X_C is the capacitive reactance. X_L and X_C are given by the formulas:

$$X_L = 2 \pi f L$$

$$X_C = 1 / (2 \pi f C)$$

- The phase angle (ϕ) is the angle between the voltage and the current in the circuit, measured in degrees or radians. It indicates the phase difference between the voltage and the current. It is given by the formula:

$$\phi = \tan^{-1}((X_L - X_C) / R)$$

- The power factor (pf) is the ratio of the real power to the apparent power in the circuit. It indicates how efficiently the circuit uses the power supplied

by the source. It is given by the formula:

$$\text{pf} = \cos(\)$$

- The power (P) is the rate of energy transfer in the circuit, measured in watts (W). It is given by the formula:

$$P = VI\text{pf}$$

- To determine the parameters of a series RLC circuit, one can use the following steps:
 1. Measure or obtain the values of R, L, C, V, and f.
 2. Calculate XL and XC using the formulas above.
 3. Calculate Z using the formula above.
 4. Calculate I using the formula $I = V / Z$.
 5. Calculate using the formula above.
 6. Calculate pf using the formula above.
 7. Calculate P using the formula above.
- Alternatively, one can use an instrument called an impedance bridge or a vector network analyzer to measure the impedance and the phase angle of the circuit directly, and then calculate the other parameters using the formulas above.

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7. Determination of (i) Voltage ratio (ii) polarity and (iii) efficiency by load test of a single phase Transformer

A single phase transformer is a device that transfers electrical energy from one circuit to another through electromagnetic induction. The transformer consists of two windings, called the primary and the secondary, that are linked by a magnetic core. The primary winding is connected to the input voltage source, while the secondary winding is connected to the output load.

The voltage ratio, polarity and efficiency of a single phase transformer can be determined by performing a load test, which involves applying a known voltage to the primary winding and measuring the corresponding voltage, current and power in the secondary winding. The load test can be done using the following steps:

- Connect the primary winding of the transformer to a variable voltage source, such as an autotransformer, and the secondary winding to a resistive load, such as a rheostat. Also connect an ammeter, a voltmeter and a wattmeter to the secondary circuit, as shown in the figure below.

Load test circuit diagram

- Adjust the input voltage to the rated value of the primary winding, and note down the readings of the voltmeter, ammeter and wattmeter in the secondary circuit. This is the no-load condition, where the secondary current is very small and the power factor is close to zero.
- Increase the load resistance gradually, and note down the corresponding readings of the voltmeter, ammeter and wattmeter in the secondary circuit. This is the loaded condition, where the secondary current and power factor increase as the load resistance decreases.
- Repeat the above step for different values of load resistance, until the rated current of the secondary winding is reached. Plot a graph of secondary voltage versus secondary current, and secondary power versus secondary current, using the recorded data.

The voltage ratio, polarity and efficiency of the transformer can be calculated from the load test data as follows:

- The voltage ratio is the ratio of the secondary voltage to the primary voltage, which can be obtained from the no-load condition. It is given by:

$$\text{Voltage ratio} = V_2 / V_1$$

where V_2 is the secondary voltage and V_1 is the primary voltage.

- The polarity is the relative direction of the induced voltages in the primary and secondary windings, which can be determined by the dot convention. The dot convention states that if the current enters the dotted terminal of one winding, the induced voltage in the other winding will have the same polarity at the dotted terminal. The polarity can be verified by connecting a low resistance across the secondary terminals, and observing the deflection of the ammeter in the primary circuit. If the ammeter deflects in the same direction as the no-load condition, the polarity is additive, otherwise it is subtractive. The polarity can be indicated by placing a plus sign (+) or a minus sign (-) near the dotted terminals of the windings, as shown in the figure below.

Polarity diagram

- The efficiency is the ratio of the output power to the input power, which can be obtained from the loaded condition. It is given by:

$$\text{Efficiency} = P_2 / P_1$$

where P_2 is the secondary power and P_1 is the primary power.

The primary power can be calculated from the secondary power and the losses in the transformer, which are mainly due to the copper loss and the iron loss. The copper loss is the power dissipated in the resistance of the windings, which is proportional to the square of the current. The iron loss is the power dissipated in the core due to hysteresis and eddy currents, which is constant for a given input voltage and frequency. The losses can be estimated from the load test data as follows:

- The copper loss is the difference between the input power and the output power at full load, which is the maximum current of the secondary winding. It is given by:

$$\text{Copper loss} = P_1 - P_2 \text{ (at full load)}$$

- The iron loss is the input power at no load, which is the minimum current of the secondary winding. It is given by:

$$\text{Iron loss} = P_1 \text{ (at no load)}$$

The primary power can then be calculated by adding the output power and the losses, as follows:

$$P_1 = P_2 + \text{Copper loss} + \text{Iron loss}$$

The efficiency can then be calculated by dividing the output power by the input power, as follows:

$$\text{Efficiency} = P_2 / (P_2 + \text{Copper loss} + \text{Iron loss})$$

The efficiency can also be plotted as

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Determination of efficiency of a dc shunt motor by load test

- A dc shunt motor is a type of dc motor where the field winding is connected in parallel with the armature winding. This means that the field current is independent of the armature current and the speed is relatively constant.
- The efficiency of a dc shunt motor is the ratio of the output power to the input power. It can be expressed as:

$$\eta = \frac{P_{out}}{P_{in}} = \frac{T\omega}{VI}$$

where η is the efficiency, P_{out} is the output power, P_{in} is the input power, T is the torque, ω is the angular speed, V is the terminal voltage, and I is the total current. - To determine the efficiency of a dc shunt motor by load test, the following steps are followed:

- Connect the dc shunt motor to a suitable load, such as a brake drum or a dynamometer, and measure the torque and speed of the motor.
- Measure the terminal voltage and the total current of the motor using a voltmeter and an ammeter.
- Calculate the output power of the motor using the formula: $P_{out} = T\omega$

- Calculate the input power of the motor using the formula: $P_{in} = VI$
- Calculate the efficiency of the motor using the formula: $\eta = \frac{P_{out}}{P_{in}}$
- Repeat the above steps for different values of load and plot a graph of efficiency versus output power. The maximum efficiency occurs at the point where the slope of the curve is zero.

9. To study running and speed reversal of a three phase induction motor and record speed in both directions.

- A three phase induction motor is a type of electric motor that converts alternating current (AC) power into mechanical power by creating a rotating magnetic field in the stator and inducing currents in the rotor.
- The speed of a three phase induction motor depends on the frequency of the AC supply, the number of poles in the stator, and the slip between the rotor and the stator. The speed can be expressed by the formula:

$$N = (120f / p) (1 - s)$$

where N is the speed in revolutions per minute (rpm), f is the frequency in hertz (Hz), p is the number of poles, and s is the slip.

- The slip is the difference between the synchronous speed (the speed of the rotating magnetic field) and the actual speed of the rotor. The slip is usually expressed as a percentage of the synchronous speed. The slip can be calculated by the formula:

$$s = (N_s - N) / N_s$$

where N_s is the synchronous speed in rpm, and N is the actual speed in rpm.

- The direction of rotation of a three phase induction motor is determined by the phase sequence of the AC supply. If the phase sequence is R-Y-B, the motor rotates in the clockwise direction. If the phase sequence is reversed to B-Y-R, the motor rotates in the anti-clockwise direction.
- To study the running and speed reversal of a three phase induction motor, the following steps are followed:
 - Connect the three phase induction motor to a three phase AC supply through a star-delta starter and a reversing switch. The star-delta starter is used to reduce the starting current and the reversing switch is used to change the phase sequence of the supply.
 - Switch on the supply and observe the direction of rotation of the motor. Note down the speed of the motor using a tachometer or a speedometer.

- Switch off the supply and change the position of the reversing switch. Switch on the supply again and observe the direction of rotation of the motor. Note down the speed of the motor using a tachometer or a speedometer.
- Compare the speeds of the motor in both directions and verify that they are equal or nearly equal. Also verify that the direction of rotation is reversed by changing the phase sequence of the supply.
- Repeat the experiment for different values of frequency and number of poles and record the speed of the motor in both directions. Plot the speed versus frequency and speed versus number of poles graphs and observe the trends.

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10. Demonstration of cut-out sections of machines: dc machine, three phase induction machine, single-phase induction machine and synchronous machine.

DC Machine

- A DC machine is a device that converts electrical energy into mechanical energy or vice versa.
- It consists of two main parts: the stator and the rotor.
- The stator is the stationary part that contains the field windings, which produce a magnetic field when energized by a DC source.
- The rotor is the rotating part that contains the armature windings, which carry the current induced by the magnetic field.
- The armature windings are connected to a commutator, which is a cylindrical device that consists of segments of copper bars insulated from each other.
- The commutator rotates with the rotor and allows the current to flow in the same direction in the external circuit by changing the connections of the armature windings to the brushes.
- The brushes are stationary conductors that make contact with the commutator segments and transfer the current to or from the external circuit.
- The direction of rotation of the rotor depends on the direction of the current in the armature windings and the polarity of the field windings.
- A DC machine can operate as a motor or a generator depending on the direction of the power flow.

DC Machine

Three Phase Induction Machine

- A three phase induction machine is a device that converts electrical energy into mechanical energy or vice versa using the principle of electromagnetic induction.
- It consists of two main parts: the stator and the rotor.
- The stator is the stationary part that contains the stator windings, which are connected to a three phase AC supply.
- The stator windings produce a rotating magnetic field that rotates at a synchronous speed, which depends on the frequency of the supply and the number of poles of the machine.
- The rotor is the rotating part that contains the rotor windings, which are either short-circuited or connected to an external resistance.
- The rotor windings are either wound or embedded in the rotor core, which is made of laminated steel to reduce eddy current losses.
- The rotor windings are either of the squirrel cage type or the wound type.
- The squirrel cage type has bars of copper or aluminum that are short-circuited at both ends by end rings, forming a cage-like structure.
- The wound type has coils of insulated wire that are connected to slip rings, which allow the connection of an external resistance or a DC source.
- The rotating magnetic field induces an electromotive force (emf) in the rotor windings, which causes a current to flow in them.
- The current in the rotor windings interacts with the magnetic field and produces a torque that causes the rotor to rotate.
- The rotor speed is less than the synchronous speed, and the difference is called the slip.
- The slip determines the amount of power transferred from the stator to the rotor and the torque developed by the machine.
- A three phase induction machine can operate as a motor or a generator depending on the direction of the power flow and the slip.

Three Phase Induction Machine

Single Phase Induction Machine

- A single phase induction machine is a device that converts electrical energy into mechanical energy or vice versa using the principle of electromagnetic induction.
- It consists of two main parts: the stator and the rotor.
- The stator is the stationary part that contains the stator winding, which is connected to a single phase AC supply.
- The stator winding produces an alternating magnetic field that oscillates along a fixed axis, rather than rotating around it.
- The rotor is the rotating part that contains the rotor winding, which is either short-circuited or connected to an external resistance.
- The rotor winding is either of the squirrel cage type or the wound type, similar to the three phase induction machine.

- The alternating magnetic field induces an emf in the rotor winding, which causes a current to flow in it.
- The current in the rotor winding interacts with the magnetic field and produces a torque that causes the rotor to rotate.
- However, the torque is not constant and varies with the position of the rotor, resulting in a pulsating

Experiments available on virtual lab

- Virtual lab is a platform that allows users to perform experiments online without the need of physical equipment or facilities.
- Virtual lab can simulate various scientific phenomena and processes, such as physics, chemistry, biology, engineering, etc.
- Virtual lab can provide interactive feedback, guidance, and assessment to the users, as well as record and store their data and results.
- Some of the experiments available on virtual lab are:
 - **Physics:**
 - * Projectile motion
 - * Pendulum
 - * Newton's laws of motion
 - * Optics
 - * Electricity and magnetism
 - * Waves and sound
 - * Thermodynamics
 - * Quantum mechanics
 - **Chemistry:**
 - * Acid-base titration
 - * Chemical equilibrium
 - * Electrochemistry
 - * Organic synthesis
 - * Spectroscopy
 - * Chromatography
 - * Molecular structure
 - * Reaction kinetics
 - **Biology:**
 - * Cell structure and function
 - * DNA extraction and analysis
 - * Enzyme activity
 - * Photosynthesis and respiration
 - * Genetics and inheritance
 - * Microbiology
 - * Ecology
 - * Evolution

- **Engineering:**
 - * Circuit design and analysis
 - * Control systems
 - * Fluid mechanics
 - * Heat transfer
 - * Material properties and testing
 - * Mechanical vibrations
 - * Robotics
 - * Signal processing

Kirchhoff's Laws

Kirchhoff's laws are a set of two laws that describe how current and voltage behave in electrical circuits. They are based on the principles of conservation of charge and energy, and they can be used to analyze complex circuits with multiple loops and branches.

Kirchhoff's Current Law (KCL)

Kirchhoff's current law, also known as Kirchhoff's first law or Kirchhoff's junction rule, states that the sum of currents entering a node (or junction) in a circuit is equal to the sum of currents leaving that node. In other words, the algebraic sum of currents at any node is zero. This law implies that charge is conserved at any point in a circuit, and that current does not accumulate or disappear at any node.

Mathematically, Kirchhoff's current law can be written as:

$$\sum_{k=1}^n I_k = 0$$

where I_k is the current of the k -th branch connected to the node, and n is the number of branches. The sign of the current depends on the direction of the current flow: positive if the current enters the node, and negative if the current leaves the node.

For example, consider the following circuit with four branches and one node:

KCL example

Applying Kirchhoff's current law to the node, we get:

$$I_1 + I_2 - I_3 - I_4 = 0$$

This equation can be used to find the value of any unknown current in the circuit, given the values of the other currents.

Kirchhoff's Voltage Law (KVL)

Kirchhoff's voltage law, also known as Kirchhoff's second law or Kirchhoff's loop rule, states that the sum of voltages around any closed loop in a circuit is zero. In other words, the algebraic sum of the potential differences across all the elements in a loop is zero. This law implies that energy is conserved in a circuit, and that the voltage drop across a loop is equal to the voltage rise across the loop.

Mathematically, Kirchhoff's voltage law can be written as:

$$\sum_{k=1}^n V_k = 0$$

where V_k is the voltage of the k -th element in the loop, and n is the number of elements. The sign of the voltage depends on the direction of the loop traversal: positive if the loop goes from the negative to the positive terminal of the element, and negative if the loop goes from the positive to the negative terminal of the element.

For example, consider the following circuit with three elements and one loop:

KVL example

Applying Kirchhoff's voltage law to the loop, we get:

$$V_1 - V_2 - V_3 = 0$$

This equation can be used to find the value of any unknown voltage in the circuit, given the values of the other voltages.

Virtual Lab

Virtual Lab is an online platform that provides remote access to various experiments in science and engineering disciplines. It is developed by Amrita Vishwa Vidyapeetham in collaboration with other institutions and supported by the Ministry of Education, Government of India. The main objectives of Virtual Lab are:

- To enhance the learning outcomes of students by providing them with hands-on experience and interactive feedback.
- To overcome the limitations of physical labs such as lack of resources, equipment, space, time, and safety.
- To enable students from different locations and backgrounds to access the same quality of education and research opportunities.
- To promote inquiry-based learning and stimulate curiosity and creativity among students.

The link provided by the user is for a specific experiment in the Virtual Lab of Biotechnology, which is one of the 12 disciplines covered by the platform. The experiment is about the isolation of genomic DNA from animal tissue. The link contains the following sections:

- **Theory:** This section explains the basic concepts and principles behind the experiment, such as the structure and function of DNA, the sources and types of animal tissue, the methods and steps of DNA extraction, and the applications and importance of DNA isolation.
- **Procedure:** This section provides a detailed and step-by-step guide on how to perform the experiment using the virtual simulation. It also includes animations, videos, and images to illustrate the process and the expected results.
- **Self Evaluation:** This section contains multiple-choice questions and answers to test the user's understanding and knowledge of the experiment. It also provides feedback and explanations for each question.
- **Simulator:** This section allows the user to interact with the virtual lab environment and conduct the experiment by following the instructions and using the tools and materials provided. The user can also view the results and compare them with the expected outcomes.
- **Assignment:** This section provides some additional questions and tasks for the user to further explore and apply the concepts and skills learned from the experiment. It also suggests some references and resources for further reading and research.
- **Reference:** This section lists some relevant and reliable sources of information that the user can consult for more details and background on the experiment and the topic. It includes books, journals, websites, and videos.

2. Thevenin Theorem

- Thevenin's theorem is a method of simplifying any linear circuit, regardless of its complexity, to an equivalent circuit with a single voltage source and a series resistance .
- Thevenin's theorem can be applied to both AC and DC circuits.
- Thevenin's theorem can be used to make circuit analysis easier and to study a circuit's initial-condition and steady-state response.
- Thevenin's theorem and its dual, Norton's theorem, are widely used in circuit theory and engineering.

Steps to apply Thevenin's theorem

- To apply Thevenin's theorem to any two-terminal linear circuit, we need to find the Thevenin's equivalent circuit, which consists of a voltage source V_{th} in series with a resistance R_{th} .
- The steps to find the Thevenin's equivalent circuit are :

- Step 1: Remove the load resistor (the resistor connected to the terminals of interest) and replace it with an open circuit.
- Step 2: Calculate the Thevenin voltage (V_{th}), which is the voltage across the open circuit.
- Step 3: Calculate the Thevenin resistance (R_{th}), which is the equivalent resistance seen from the terminals of the open circuit, with all independent sources turned off (voltage sources replaced by short circuits and current sources replaced by open circuits).
- Step 4: Connect the Thevenin voltage source and the Thevenin resistance in series to form the Thevenin's equivalent circuit.

Example of applying Thevenin's theorem

- Consider the following circuit, where we want to find the Thevenin's equivalent circuit across the terminals A and B.

Circuit diagram

- Step 1: Remove the load resistor R_3 and replace it with an open circuit.

Circuit diagram with open circuit

- Step 2: Calculate the Thevenin voltage (V_{th}), which is the voltage across the open circuit. We can use the voltage divider rule to find V_{th} :

$$V_{th} = V_1 * (R_2 / (R_1 + R_2)) = 10 \text{ V} * (3 \Omega / (2 \Omega + 3 \Omega)) = 6 \text{ V}$$

- Step 3: Calculate the Thevenin resistance (R_{th}), which is the equivalent resistance seen from the terminals of the open circuit, with all independent sources turned off. In this case, we replace V_1 with a short circuit.

Circuit diagram with short circuit

$$R_{th} = R_1 || R_2 = (R_1 * R_2) / (R_1 + R_2) = (2 \Omega * 3 \Omega) / (2 \Omega + 3 \Omega) = 1.2 \Omega$$

- Step 4: Connect the Thevenin voltage source and the Thevenin resistance in series to form the Thevenin's equivalent circuit.

Thevenin's equivalent circuit

- The Thevenin's equivalent circuit can be used to analyze the behavior of the original circuit with any load resistor connected to the terminals A and B. For example, if we reconnect $R_3 = 4 \Omega$, we can find the current through R_3 as:

$$I_3 = V_{th} / (R_{th} + R_3) = 6 \text{ V} / (1.2 \Omega + 4 \Omega) = 1.2 \text{ A}$$

- This is the same current that would flow through R_3 in the original circuit.

Virtual Lab

- Virtual Lab is an online platform that provides remote access to various experiments in science and engineering disciplines.
- Virtual Lab is developed by Amrita Vishwa Vidyapeetham in collaboration with other institutions and supported by the Ministry of Education, Government of India.
- Virtual Lab aims to enhance the quality of education by providing students with a realistic and interactive laboratory experience, regardless of their location, time, and resources.
- Virtual Lab consists of simulations, animations, videos, and graphics that mimic the real-world scenarios and phenomena of the experiments.
- Virtual Lab covers various domains such as physics, chemistry, biology, biotechnology, computer science, electrical engineering, mechanical engineering, and civil engineering.
- Virtual Lab allows students to perform the experiments at their own pace, repeat them as many times as they want, and explore different parameters and outcomes.
- Virtual Lab also provides feedback, assessment, and guidance to help students learn the concepts and skills involved in the experiments.
- Virtual Lab is accessible through the website <https://vlab.amrita.edu/> or the mobile app Amrita Online Lab.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of RLC series resonance. Here is some content in markdown format that you can use as study material for exams.

RLC series resonance

- RLC series resonance is a phenomenon that occurs in a circuit containing a resistor (R), an inductor (L), and a capacitor (C) connected in series.
- When the circuit is driven by an alternating voltage source with a frequency that matches the natural frequency of the circuit, the circuit exhibits maximum current and minimum impedance.
- The natural frequency of the circuit is given by the formula:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

- At this frequency, the reactance of the inductor and the capacitor cancel each other, and the circuit behaves like a pure resistor with a resistance equal to R.
- The current in the circuit is in phase with the voltage source, and the voltage across each component is proportional to its impedance.
- The voltage across the resistor is equal to the source voltage, the voltage across the inductor leads the current by 90 degrees, and the voltage across

the capacitor lags the current by 90 degrees.

- The phasor diagram of the circuit at resonance is shown below:

RLC series resonance phasor diagram

- The quality factor (Q) of the circuit is a measure of how sharp the resonance is, and is given by the formula:

$$Q = \frac{f_0}{\Delta f} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

- The quality factor indicates how much the current in the circuit increases at resonance compared to the current at other frequencies. A higher Q means a narrower bandwidth and a more selective circuit.
- The bandwidth (Δf) of the circuit is the range of frequencies around the resonant frequency where the current is at least 70.7% of the maximum current. It is given by the formula:

$$\Delta f = \frac{f_0}{Q} = \frac{R}{2\pi L}$$

- The power factor (PF) of the circuit is the ratio of the real power (P) to the apparent power (S) in the circuit. It is given by the formula:

$$PF = \frac{P}{S} = \frac{V_R I}{V_S I} = \frac{R}{Z}$$

- The power factor indicates how efficiently the circuit converts the input voltage to useful power. A higher PF means less reactive power and less power loss in the circuit.
- At resonance, the power factor of the circuit is 1, meaning that the circuit is purely resistive and there is no reactive power. The real power is equal to the apparent power, and is given by the formula:

$$P = S = V_S I = I^2 R$$

- The following graph shows the variation of current, impedance, and power factor with frequency in a series RLC circuit:

RLC series resonance graph

- The graph shows that the current is maximum and the impedance and power factor are minimum at the resonant frequency. The current decreases and the impedance and power factor increase as the frequency deviates from the resonant frequency.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of virtual lab. Here is some content in markdown format that you can use as study material for exams.

Virtual Lab

- A virtual lab is a computer-based simulation of a real laboratory environment that allows users to perform experiments, collect data, and analyze results without the need for physical equipment, materials, or space.
- Virtual labs can be used for various purposes, such as education, research, training, testing, and entertainment.
- Virtual labs can offer many benefits, such as:
 - Reducing the cost, time, and risk of conducting experiments in real labs.
 - Enhancing the accessibility, availability, and diversity of learning resources and experiences for students and teachers.
 - Enabling the exploration, manipulation, and visualization of phenomena that are otherwise difficult, dangerous, or impossible to observe or control in real labs.
 - Supporting the development of scientific inquiry, problem-solving, and critical thinking skills among learners.
 - Promoting the collaboration, communication, and feedback among peers, instructors, and experts in virtual communities.
- Virtual labs can also have some limitations, such as:
 - Requiring adequate technological infrastructure, software, and internet connectivity to run smoothly and reliably.
 - Lacking the authenticity, realism, and physicality of real labs, which may affect the motivation, engagement, and learning outcomes of some users.
 - Posing ethical, legal, and social issues related to the ownership, privacy, security, and quality of the virtual lab content and data.
 - Demanding new pedagogical approaches, strategies, and assessment methods to effectively integrate virtual labs into the curriculum and instruction.
- A virtual lab consists of three main components: the simulation, the interface, and the interaction.
 - The simulation is the core of the virtual lab, which models the behavior and dynamics of the real system or phenomenon under study, using mathematical equations, algorithms, and data.
 - The interface is the graphical representation of the virtual lab, which displays the simulation output, the experimental settings, and the user controls, using images, animations, sounds, and text.
 - The interaction is the communication between the user and the virtual lab, which allows the user to manipulate the simulation input, observe the simulation output, and receive feedback, using devices such as mouse, keyboard, touch screen, microphone, or headset.
- A virtual lab can be classified into different types, based on the level of realism, interactivity, and complexity of the simulation, such as:
 - Static virtual labs, which present fixed or predetermined scenarios and outcomes, with no or minimal user input or feedback.

- Dynamic virtual labs, which generate variable or random scenarios and outcomes, with moderate user input and feedback.
- Adaptive virtual labs, which adjust the scenarios and outcomes according to the user's performance, preferences, and needs, with high user input and feedback.
- Intelligent virtual labs, which incorporate artificial intelligence techniques, such as machine learning, natural language processing, and computer vision, to create personalized, adaptive, and immersive scenarios and outcomes, with advanced user input and feedback.
- A virtual lab can be accessed through different platforms, such as:
 - Web-based virtual labs, which run on web browsers and require internet connection.
 - Desktop-based virtual labs, which run on personal computers and require software installation.
 - Mobile-based virtual labs, which run on smartphones or tablets and require app download.
 - Virtual reality-based virtual labs, which run on head-mounted displays and require special hardware and software.
- A virtual lab can be developed using different tools, such as:
 - Programming languages, such as Java, Python, C++, or HTML, which require coding skills and software development environments.
 - Authoring tools, such as Adobe Flash, Unity, or LabVIEW, which provide graphical user interfaces and pre-built components to create virtual labs.
 - Learning management systems, such as Moodle, Blackboard, or Canvas, which provide online platforms and features to host and manage virtual labs.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some information that I found from the web:

4. Measurement of power in 3- phase circuit by two wattmeter method and determination of its power factor for star as well as delta connected load.

- The two wattmeter method is a technique for measuring the total power in a three-phase circuit using two wattmeters.
- The current coils of the two wattmeters are connected in series with any two line conductors, and the potential coils of each wattmeter are connected to the third line conductor .
- The total power is given by the sum of the two wattmeter readings: $P = P_1 + P_2$
- The power factor is given by the ratio of the total power to the apparent

power: $PF = P / (\sqrt{3} * V * I)$, where V is the line voltage and I is the line current.

- For a balanced three-phase load, the power factor can also be calculated from the difference of the two wattmeter readings: $PF = \cos(\phi) = (P1 + P2) / (\sqrt{3} * (P1 - P2))$
- For a star-connected load, the line voltage is equal to the phase voltage times $\sqrt{3}$, and the line current is equal to the phase current: $V = \sqrt{3} * V_p$, $I = I_p$
- For a delta-connected load, the line voltage is equal to the phase voltage, and the line current is equal to the phase current times $\sqrt{3}$: $V = V_p$, $I = \sqrt{3} * I_p$
- The two wattmeter method can be used for both balanced and unbalanced loads, but the power factor formula is only valid for balanced loads .

Hello, I am Sydney, your AI assistant. I can help you with your topic.

Virtual Lab: Dreamweaver Measurement

- A virtual lab is a simulated environment that allows students to perform experiments or activities online without the need for physical equipment or facilities .
- Dreamweaver is a web development software that enables users to create, edit, and manage websites and web applications.
- Measurement is the process of determining the size, quantity, or degree of something using a standard unit or system.

Objectives

- To learn about the different types of measuring devices used in electrical engineering, such as ammeter, voltmeter, wattmeter, and power factor meter .
- To understand the principles and techniques of measuring electrical power, voltage, current, and power factor in various circuits .
- To practice using Dreamweaver to create and modify web pages that display the results of the virtual lab experiments.

Procedure

- Access the virtual lab link: <http://vp-dei.vlabs.ac.in/Dreamweaver/measurement.html>
- Read the instructions and theory sections to familiarize yourself with the concepts and terms related to the virtual lab .
- Choose one of the four experiments from the menu: power measurement, voltage measurement, current measurement, or power factor measurement .
- Follow the steps given in the experiment page to set up the circuit, connect the measuring devices, and observe the readings .

- Record the data and calculations in a table or a spreadsheet .
- Use Dreamweaver to create a web page that displays the data and calculations, as well as a summary and conclusion of the experiment.
- Save and publish the web page using Dreamweaver's built-in features.
- Review and evaluate your web page for accuracy, clarity, and design.

References

: **Virtual Labs - DEI - vlab.co.in** Virtual Labs - DEI

Options for Virtual Labs and Simulations for Laboratory Based Courses

...

Measurement Virtual Labs - warrencountyschools.org

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic of 5. Determination of parameters of ac single phase series RLC circuit.

5. Determination of parameters of ac single phase series RLC circuit

- An ac single phase series RLC circuit consists of a resistor (R), an inductor (L), and a capacitor (C) connected in series to an ac voltage source (V).
- The current (I) in the circuit is the same for all the components and is given by Ohm's law: $I = V/Z$, where Z is the total impedance of the circuit.
- The impedance Z is a complex quantity that depends on the frequency (f) of the ac source and the values of R, L, and C. It can be written as: $Z = R + jX$, where j is the imaginary unit and X is the total reactance of the circuit.
- The reactance X is the sum of the inductive reactance (XL) and the capacitive reactance (XC). $X_L = 2\pi fL$ and $X_C = 1/(2\pi fC)$.
- The impedance Z can also be expressed in polar form as: $Z = |Z| \angle \theta$, where |Z| is the magnitude of Z and θ is the phase angle between the voltage and the current. $|Z| = \sqrt{R^2 + X^2}$ and $\theta = \tan^{-1}(X/R)$.
- The power factor (pf) of the circuit is the cosine of the phase angle: $\text{pf} = \cos(\theta)$. It indicates how efficiently the circuit converts the ac voltage into useful power. A pf of 1 means that the voltage and the current are in phase and the circuit is purely resistive. A pf of 0 means that the voltage and the current are 90 degrees out of phase and the circuit is purely reactive.
- The parameters of the ac single phase series RLC circuit can be determined by measuring the voltage, the current, and the power factor of the circuit. Alternatively, they can be calculated by using the formulas given above.

Virtual Lab

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- Virtual Lab aims to enhance the quality of education by providing students with the opportunity to perform experiments in a virtual environment, without the need for physical infrastructure, equipment, or supervision.
- Virtual Lab consists of simulations, animations, videos, and interactive elements that mimic the real-world scenarios and phenomena of the experiments.
- Virtual Lab covers various domains such as physics, chemistry, biology, biotechnology, computer science, electrical engineering, mechanical engineering, civil engineering, and chemical engineering.
- Virtual Lab also provides assessment tools, feedback mechanisms, and learning resources to help students evaluate their performance and understanding of the experiments.
- Virtual Lab is accessible through the website <https://vlab.amrita.edu/> or the mobile app Amrita Online Lab.

Hello, I am Sydney, your AI assistant. I can help you with your topic.

6. To observe the B-H loop of a ferromagnetic material in CRO.

- A B-H loop is a graphical representation of the relationship between the magnetic flux density (B) and the magnetic field intensity (H) of a ferromagnetic material.
- A ferromagnetic material is one that can be magnetized by an external magnetic field and retain some magnetization after the field is removed.
- A CRO (cathode ray oscilloscope) is an instrument that can display the variation of an electrical signal on a screen.
- To observe the B-H loop of a ferromagnetic material in CRO, the following steps are followed:
 1. A ferromagnetic material, such as an iron ring, is wound with two coils: a primary coil and a secondary coil. The primary coil is connected to an AC source, which provides the varying magnetic field H. The secondary coil is connected to a galvanometer, which measures the induced EMF due to the changing magnetic flux B.
 2. The CRO is set up with the X-axis input connected to the AC source and the Y-axis input connected to the galvanometer. The CRO is adjusted to obtain a stable and clear display on the screen.
 3. As the AC source is varied, the magnetic field H changes in magnitude and direction. This causes the magnetic flux B to change accordingly,

inducing an EMF in the secondary coil. The EMF is proportional to the rate of change of B , and its polarity depends on the direction of B .

4. The CRO displays the variation of B (Y-axis) with respect to H (X-axis) as a loop-shaped curve on the screen. This is the B - H loop of the ferromagnetic material.
5. The shape and size of the B - H loop depend on the properties of the ferromagnetic material, such as its coercivity, remanence, saturation, and hysteresis. These parameters can be determined from the B - H loop as follows:
 - Coercivity: The value of H required to reduce B to zero. It is a measure of the resistance of the material to demagnetization.
 - Remanence: The value of B remaining when H is reduced to zero. It is a measure of the residual magnetization of the material.
 - Saturation: The maximum value of B that the material can attain for a given H . It is a measure of the magnetic capacity of the material.
 - Hysteresis: The area enclosed by the B - H loop. It is a measure of the energy loss due to the reversal of magnetization of the material.
6. The B - H loop of a ferromagnetic material in CRO can be used to study the magnetic behavior and characteristics of the material, and to compare different materials for their suitability in various applications.

Virtual Lab

Virtual Lab is an online platform that provides remote access to various laboratories in science and engineering disciplines. It is developed by Amrita Vishwa Vidyapeetham in collaboration with other institutions and supported by the Ministry of Education, Government of India. Virtual Lab aims to enhance the quality of education and research by providing students and teachers with interactive simulations, animations, videos and assessments of real-world experiments.

Some of the objectives of Virtual Lab are:

- To provide a complete learning environment for performing experiments that are otherwise not feasible or accessible due to cost, safety, time or location constraints.
- To supplement the theoretical concepts taught in the classroom with practical demonstrations and applications.
- To develop the skills of observation, analysis, interpretation and inference among the learners.
- To foster curiosity, creativity and innovation among the learners.
- To enable collaborative learning and sharing of resources among the learners and teachers across different institutions and regions.

The link provided in the topic is for a specific experiment in the Virtual Lab of Physical Sciences, which is one of the 12 domains covered by the platform. The experiment is about the **Determination of the Focal Length of a Concave Mirror**. It involves the following steps:

- Setting up the apparatus, which consists of a concave mirror, an optical bench, a light source, a screen and a meter scale.
- Adjusting the position of the light source and the screen to obtain a sharp image of the object on the screen.
- Measuring the distance between the mirror and the object (u) and the distance between the mirror and the image (v) using the meter scale.
- Repeating the procedure for different values of u and v and recording the observations in a table.
- Plotting a graph of $1/u$ versus $1/v$ and finding the slope and the intercept of the line.
- Using the mirror formula $1/f = 1/u + 1/v$ and the relation $f = -v/u$ to calculate the focal length of the mirror from the slope and the intercept of the graph.
- Comparing the values of the focal length obtained from the graph and the formula and finding the percentage error.

The experiment also provides the following features:

- A theory section that explains the concepts and principles behind the experiment.
- A procedure section that gives the detailed steps and instructions for performing the experiment.
- A video section that shows the actual experiment being conducted in a laboratory.
- A simulation section that allows the user to perform the experiment virtually by manipulating the parameters and observing the results.
- A self-evaluation section that tests the user's understanding of the experiment with multiple choice questions and feedback.
- A reference section that provides additional resources and links for further learning.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some information that I found from the web.

Determination of the efficiency of a dc motor by loss summation method (Swinburne's test)

- The efficiency of a dc motor is defined as the ratio of output (mechanical) power to input (electrical) power.
- The efficiency of a dc motor can be measured by direct or indirect methods.
- The direct method requires measuring the input power by a power analyzer

and the output power by a speed and torque meter.

- The indirect method, also known as the loss summation method or Swinburne's test, involves determining the losses of the dc motor without loading it.
- The losses of a dc motor can be classified into constant losses and variable losses.
- The constant losses are the losses that do not depend on the load, such as core loss, friction loss, and windage loss.
- The variable losses are the losses that depend on the load, such as copper loss in the armature and the field windings.
- The Swinburne's test is performed by running the dc motor at no load and rated speed, and measuring the no-load current and voltage.
- The no-load input power is equal to the sum of the constant losses and the shunt field copper loss.
- The armature copper loss can be calculated by multiplying the armature resistance and the square of the rated current.
- The total losses can be obtained by adding the no-load input power and the armature copper loss.
- The efficiency of the dc motor can be calculated by subtracting the total losses from the rated output power and dividing by the rated output power.
- The Swinburne's test has the advantages of convenience, economy, and accuracy, as it does not require loading the motor or measuring the output power.
- The Swinburne's test has the limitations of being applicable only to shunt and compound motors, and not accounting for the temperature rise and stray load losses that occur at full load.

: Testing of DC Machines (DC Motor and DC Generator) - StudyElectrical.Com

Efficiency of DC Motor - ElectricalWorkbook

IEC 60034-2-1 standard on efficiency measurement methods for low ... - ABB

Course Outcomes:

- Course outcomes are statements that describe what students are expected to know and be able to do by the end of a course.
- Course outcomes are aligned with the course objectives, which are derived from the program outcomes and the institutional mission and vision.
- Course outcomes are measurable, observable, and achievable within the scope and duration of the course.
- Course outcomes are written in terms of student learning, using action verbs that indicate the level of cognitive, affective, or psychomotor skills required.
- Course outcomes are communicated to the students at the beginning of the course and throughout the course delivery.
- Course outcomes are assessed using appropriate methods and tools, such

as exams, assignments, projects, portfolios, presentations, etc.

- Course outcomes are used to evaluate the effectiveness of the course design, delivery, and improvement, as well as the student achievement and satisfaction.

Course Outcome (CO) Bloom's Level

- A course outcome (CO) is a statement that describes what a learner should be able to do or demonstrate after completing a course or a unit of instruction.
- A CO should be specific, measurable, achievable, relevant, and time-bound (SMART).
- A CO should also align with the program outcomes (POs) and the institutional outcomes (IOs) of the educational institution.
- A CO should be written using an action verb that indicates the level of cognitive skill or ability required by the learner to achieve the outcome.
- Bloom's taxonomy is a framework that classifies different levels of cognitive skills or abilities, from lower-order to higher-order, as follows:
 - Remember: The learner can recall or recognize facts, terms, concepts, or procedures.
 - Understand: The learner can explain, interpret, summarize, or paraphrase information or ideas.
 - Apply: The learner can use or apply knowledge or skills to solve problems or perform tasks.
 - Analyze: The learner can break down information or ideas into components or relationships, or identify patterns or trends.
 - Evaluate: The learner can judge, critique, or assess the quality, value, or effectiveness of information, ideas, or arguments.
 - Create: The learner can generate, produce, or synthesize new or original information, ideas, or products.
- Bloom's level of a CO indicates the level of cognitive skill or ability that the learner is expected to demonstrate after completing the course or the unit of instruction.
- Bloom's level of a CO should be consistent with the level of difficulty and complexity of the course content and the assessment methods.
- Bloom's level of a CO should also match the level of the POs and the IOs that the CO supports or contributes to.
- Bloom's level of a CO can be determined by using a verb list or a verb chart that maps different action verbs to different levels of Bloom's taxonomy.
- Examples of COs with different Bloom's levels are:

- CO1: Define the basic concepts and principles of accounting. (Remember)
- CO2: Explain the accounting cycle and the financial statements. (Understand)
- CO3: Apply the accounting rules and standards to record business transactions and prepare financial reports. (Apply)
- CO4: Analyze the financial performance and position of a company using ratio analysis and common-size analysis. (Analyze)
- CO5: Evaluate the ethical and social implications of accounting decisions and practices. (Evaluate)
- CO6: Create a budget and a cash flow forecast for a small business. (Create)

At the end of this course, the students should be able to:

- Explain the basic concepts and principles of artificial intelligence, such as agents, search, knowledge representation, reasoning, planning, learning, natural language processing, computer vision, and robotics.
- Apply various artificial intelligence techniques to solve problems, such as heuristic search, constraint satisfaction, logic programming, probabilistic inference, decision making, neural networks, and machine learning algorithms.
- Evaluate the strengths and limitations of different artificial intelligence methods and tools, and compare their performance and applicability to different domains and scenarios.
- Design and implement simple artificial intelligence systems using Python and relevant libraries, such as NumPy, SciPy, scikit-learn, TensorFlow, PyTorch, and OpenAI Gym.
- Demonstrate ethical awareness and social responsibility when developing and using artificial intelligence systems, and consider the potential impacts and risks of artificial intelligence on society, economy, environment, and human rights.

CO 1 Conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits. K3

- KVL stands for Kirchhoff's Voltage Law, which states that the algebraic sum of the voltages around any closed loop in a circuit is zero.
- KCL stands for Kirchhoff's Current Law, which states that the algebraic sum of the currents entering and leaving any node in a circuit is zero.
- Network theorems are mathematical tools that can be used to simplify and analyze complex DC circuits. Some of the common network theorems are:

- Superposition theorem: The voltage or current in any branch of a linear circuit is equal to the algebraic sum of the voltages or currents produced by each source acting alone, with all other sources replaced by their internal resistances.
- Thevenin's theorem: Any linear circuit with voltage and current sources and only resistances can be replaced by an equivalent circuit consisting of a single voltage source in series with a single resistance connected across the load.
- Norton's theorem: Any linear circuit with voltage and current sources and only resistances can be replaced by an equivalent circuit consisting of a single current source in parallel with a single resistance connected across the load.
- Maximum power transfer theorem: The maximum power is transferred from a source to a load when the load resistance is equal to the source resistance.
- Reciprocity theorem: The current in any branch of a bilateral network due to a single source is equal to the current in the branch interchanged with the source.
- To conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits, the following steps are suggested:
 - Design and construct a DC circuit with resistors, voltage and current sources, switches, meters, and other components as required.
 - Apply KVL and KCL to verify the voltages and currents in the circuit. Use a multimeter to measure the actual values and compare them with the theoretical values. Note any discrepancies and sources of error.
 - Apply the network theorems to simplify and analyze the circuit. Use a multimeter to measure the voltages and currents in the equivalent circuit and compare them with the original circuit. Note any discrepancies and sources of error.
 - Vary the values of the resistors, sources, and load and observe the effects on the circuit behavior. Record the data and plot the graphs as needed.
 - Write a report summarizing the objectives, procedures, results, and conclusions of the experiment. Include the circuit diagrams, calculations, tables, graphs, and error analysis.

CO 2 Demonstrate the behavior of AC circuits connected to single phase AC supply and measure power in single phase as well as three phase electrical circuits. K4

- An AC circuit is a circuit that carries alternating current, which is the current that changes its direction and magnitude periodically.
- A single phase AC supply is a type of AC supply that has only one sinusoidal voltage and current waveform, with a constant frequency and amplitude.
- A single phase AC circuit can be connected to a single phase AC supply in different ways, such as series, parallel, or series-parallel combinations, depending on the load impedance and the desired voltage and current levels.
- The behavior of a single phase AC circuit can be analyzed using the following concepts:
 - Phasor diagrams: These are graphical representations of the voltage and current waveforms as rotating vectors, with their magnitudes and angles relative to a reference axis.
 - Complex numbers: These are mathematical expressions of the form $a + jb$, where a and b are real numbers, and j is the imaginary unit that satisfies $j^2 = -1$. Complex numbers can be used to represent phasors in rectangular or polar forms, and can be manipulated using algebraic rules.
 - Impedance: This is the ratio of the phasor voltage to the phasor current in an AC circuit, and it is a complex number that has a real part (resistance) and an imaginary part (reactance). Impedance can be calculated using Ohm's law, Kirchhoff's laws, or equivalent circuits.
 - Power factor: This is the cosine of the angle between the phasor voltage and the phasor current in an AC circuit, and it measures how efficiently the circuit converts the input power to the output power. Power factor can be improved by adding capacitors or inductors to the circuit to cancel the reactive power.
- The power in a single phase AC circuit can be measured using a wattmeter, which is a device that measures the product of the voltage and the current in an AC circuit. A wattmeter consists of two coils, one connected in series with the load (current coil), and the other connected across the supply (voltage coil). The wattmeter reading is proportional to the power factor of the circuit, and it can be corrected by multiplying it by the power factor.
- A three phase AC supply is a type of AC supply that has three sinusoidal voltage and current waveforms, with the same frequency and amplitude, but with a phase difference of 120 degrees between each pair. A three

phase AC supply can be connected to a three phase AC circuit in two ways, either star (Y) or delta (Δ), depending on the load impedance and the desired voltage and current levels.

- The power in a three phase AC circuit can be measured using one, two, or three wattmeters, depending on the type of connection and the availability of a neutral wire. The following formulas can be used to calculate the total power from the wattmeter readings:
 - For a balanced star connection with a neutral wire, use one wattmeter and the formula $P = W1$
 - For a balanced star connection without a neutral wire, use two wattmeters and the formula $P = W1 + W2$
 - For a balanced delta connection, use two wattmeters and the formula $P = W1 + W2$
 - For an unbalanced star or delta connection, use three wattmeters and the formula $P = W1 + W2 + W3$

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some content that you can use to study and learn from for exams.

CO 3 Perform experiment illustrating BH curve of magnetic materials. K3

- The BH curve of a magnetic material is a plot of the magnetic flux density B versus the magnetic field intensity H.
- The BH curve shows the relationship between the magnetization of the material and the applied magnetic field.
- The BH curve can be used to measure the magnetic properties of the material, such as permeability, coercivity, saturation, hysteresis, and remanence.
- To perform the experiment, you will need the following equipment and materials:
 - A solenoid coil with a variable current source and an ammeter
 - A magnetic material sample that can fit inside the coil
 - A search coil with a voltmeter
 - A magnetometer or a gaussmeter
- The steps of the experiment are as follows:
 - Connect the solenoid coil to the current source and the ammeter, and the search coil to the voltmeter.
 - Place the magnetic material sample inside the solenoid coil, and the search coil around the sample.
 - Adjust the current in the solenoid coil to zero, and record the reading of the voltmeter. This is the initial induced voltage in the search coil, V_0 .
 - Increase the current in the solenoid coil gradually, and record the readings of the ammeter and the voltmeter. The current in the

solenoid coil is proportional to the magnetic field intensity H , and the induced voltage in the search coil is proportional to the rate of change of the magnetic flux density B .

- Plot the values of B versus H on a graph paper, where B is calculated as $B = (V - V_0) / (N * A)$, where V is the induced voltage, N is the number of turns of the search coil, and A is the cross-sectional area of the sample.
- Repeat the steps from 4 to 5, but in the opposite direction, by decreasing the current in the solenoid coil gradually until it reaches zero, and then reversing the polarity of the current source. This will trace the hysteresis loop of the magnetic material.
- Measure the magnetic flux density of the sample using the magnetometer or the gaussmeter, and compare it with the value obtained from the graph. This will give you the remanence of the material, which is the residual magnetization after the applied field is removed.
- Find the value of H that corresponds to the point where B becomes zero on the graph. This will give you the coercivity of the material, which is the amount of reverse field required to demagnetize the material.
- Find the value of B that corresponds to the point where the graph becomes horizontal on both sides. This will give you the saturation of the material, which is the maximum magnetization that the material can attain.
- Find the ratio of B to H at any point on the graph. This will give you the permeability of the material, which is the measure of how easily the material can be magnetized.

CO 4 Calculate efficiency of a single phase transformer and DC machine. K4

- The efficiency of a single phase transformer is defined as the ratio of output power to input power, expressed as a percentage .
- The input and output power are measured in the same unit, either in kW or W.
- The efficiency of a single phase transformer is given by the formula :

$$\eta = \frac{P_o}{P_i} \times 100\%$$

where η is the efficiency, P_o is the output power, and P_i is the input power.

- The output power of a single phase transformer is equal to the product of the secondary voltage, the secondary current, and the power factor of the load :

$$P_o = V_s I_s \cos \phi$$

where V_s is the secondary voltage, I_s is the secondary current, and $\cos \phi$ is the power factor of the load.

- The input power of a single phase transformer is equal to the sum of the output power and the losses in the transformer :

$$P_i = P_o + P_L$$

where P_L is the total losses in the transformer.

- The losses in a single phase transformer consist of two types: core losses and copper losses .
- Core losses are caused by the alternating magnetic flux in the core, which induces eddy currents and hysteresis in the core material .
 - Eddy currents are circular currents that flow in the core due to the changing magnetic flux. They produce heat and reduce the efficiency of the transformer .
 - Hysteresis is the phenomenon of the magnetic domains in the core material aligning and realigning with the magnetic flux. This process also produces heat and reduces the efficiency of the transformer .
- Core losses are constant and independent of the load. They can be reduced by using a laminated core, which reduces the eddy currents, and by using a high permeability and low hysteresis material, such as silicon steel, for the core .
- Copper losses are caused by the resistance of the primary and secondary windings, which produce heat when current flows through them .
 - Copper losses are proportional to the square of the current and depend on the load. They can be reduced by using thicker wires, which have lower resistance, for the windings .
- Copper losses are given by the formula :

$$P_C = I_p^2 R_p + I_s^2 R_s$$

where P_C is the copper losses, I_p is the primary current, R_p is the primary resistance, I_s is the secondary current, and R_s is the secondary resistance.

- The equivalent resistance of the transformer, as seen from the primary side, is given by the formula :

$$R_e = R_p + \left(\frac{N_p}{N_s}\right)^2 R_s$$

where R_e is the equivalent resistance, N_p is the number of turns in the primary winding, and N_s is the number of turns in the secondary winding.

- The efficiency of a single phase transformer can be maximized when the core losses are equal to the copper losses .
 - This condition is called the maximum efficiency condition and it occurs at a certain load current, called the maximum efficiency current .
 - The maximum efficiency current can be calculated

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some content that you can use for your study material:

CO 5 Perform experiments on speed measurement and reversal of direction of three phase induction motor and Identify the type of DC and AC machines based on their construction. K4

- A three phase induction motor is a type of AC machine that converts electrical power into mechanical power by creating a rotating magnetic field in the stator and inducing currents in the rotor.
- The speed of a three phase induction motor depends on the frequency of the supply voltage, the number of poles in the stator, and the slip between the rotor and the stator.
- The speed of a three phase induction motor can be measured by using a tachometer, a stroboscope, or a frequency meter.
- The direction of rotation of a three phase induction motor can be reversed by interchanging any two of the three supply terminals. This changes the phase sequence of the stator currents and reverses the direction of the rotating magnetic field.
- A DC machine is a type of electrical machine that converts electrical power into mechanical power or vice versa by using a commutator and brushes to switch the direction of the current in the armature.
- A DC machine can be classified into two types based on their construction: DC generator and DC motor.
- A DC generator is a DC machine that converts mechanical power into electrical power by rotating a coil of wire in a magnetic field and inducing an emf in the coil.
- A DC motor is a DC machine that converts electrical power into mechanical power by applying a current to a coil of wire in a magnetic field and

creating a torque on the coil.

- An AC machine is a type of electrical machine that converts electrical power into mechanical power or vice versa by using alternating currents and magnetic fields.
- An AC machine can be classified into two types based on their construction: AC generator and AC motor.
- An AC generator is an AC machine that converts mechanical power into electrical power by rotating a coil of wire in a magnetic field and inducing an alternating emf in the coil.
- An AC motor is an AC machine that converts electrical power into mechanical power by applying an alternating current to a coil of wire in a magnetic field and creating a rotating magnetic field that interacts with the rotor.

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

- These are the six levels of cognitive learning according to Bloom's taxonomy, a framework for classifying educational objectives and outcomes.
- The levels are arranged in a hierarchy, from lower-order to higher-order thinking skills, and represent the degree of complexity and depth of learning.
- The levels are:
 - **K1 – Remember:** This level involves recalling facts, terms, definitions, concepts, or procedures from memory. Examples of verbs that indicate this level are: list, name, define, identify, label, recognize, recall, etc.
 - **K2 – Understand:** This level involves explaining the meaning, interpretation, or summary of information, ideas, or concepts. Examples of verbs that indicate this level are: describe, explain, paraphrase, summarize, illustrate, classify, compare, etc.
 - **K3 – Apply:** This level involves using knowledge, rules, methods, or principles to solve problems or perform tasks in new or familiar situations. Examples of verbs that indicate this level are: apply, use, demonstrate, solve, calculate, implement, etc.
 - **K4 – Analyze:** This level involves breaking down information, ideas, or concepts into parts or components, and examining the relationships, patterns, or connections among them. Examples of verbs that indicate this level are: analyze, compare, contrast, differentiate, distinguish, examine, etc.
 - **K5 – Evaluate:** This level involves making judgments, assessments, or critiques based on criteria, standards, or evidence. Examples of verbs that indicate this level are: evaluate, judge, critique, appraise,

argue, justify, etc.

- **K6 – Create:** This level involves generating, producing, or synthesizing new or original ideas, products, or solutions from existing or given information, ideas, or concepts. Examples of verbs that indicate this level are: create, design, develop, construct, compose, invent, etc.
- The levels can be used to design learning objectives, activities, and assessments that align with the intended cognitive outcomes of instruction.